
**IUTF Dataset:
Enabling CrossBorder Resource
for Analysing
the Impact of Rainfall on Urban
Transportation**

Introduction

- Motivation of Integrated Urban Traffic-Flood (IUTF) dataset:

Understanding the impact of extreme weather, particularly flooding, on urban transportation systems is critical for enhancing city resilience and traffic management. However, research and policy development are often hampered by a lack of datasets that comprehensively integrate detailed traffic dynamics, high-resolution weather information, and road network topology across multiple diverse urban environments.

- Construction of the dataset:

This open-access resource covers 40 major cities across Europe, North America, and Asia, including 21,739 sensors. The IUTF dataset uniquely combines:

1. High-resolution traffic parameters derived from over 21,700 sensors (with raw data typically at 5-minute intervals, harmonised to hourly);
2. Detailed hourly precipitation data from ERA5 reanalysis, spatially aligned with
3. The underlying road network topology for over 1 million road segments, processed from OpenStreetMap.

Introduction

Why IUTF Is Needed

- Data silos: Traffic data (such as METR-LA, PEMS-BAY) lacks a weather dimension, and meteorological data (such as ERA5) is too coarse-grained and not linked to the road network.
- Matching Barriers: encountering "triple mismatches"—coordinate mismatch, time/time zone inconsistency, and sampling granularity differences (e.g., 5-minute instantaneous flow rate vs. 1-hour cumulative precipitation).
- Lack of Multi-City Scope: Past studies have mostly focused on single cities or single countries, lacking standardized tables that can conduct comparative analyses across climates and regions.

Introduction

Data volume

- City scale: 40 core cities spanning three continents: Europe, America, and Asia.
- Traffic sensors: 21,739 (5-minute sampling rate, over 410,000 records per sensor).
- Road network: 1,067,085 road arcs extracted based on OSM (including attributes and topological relationships).
- Meteorological data: ERA5 hourly precipitation, covering 3,356 city-days.

Introduction

Contribution & Impacts

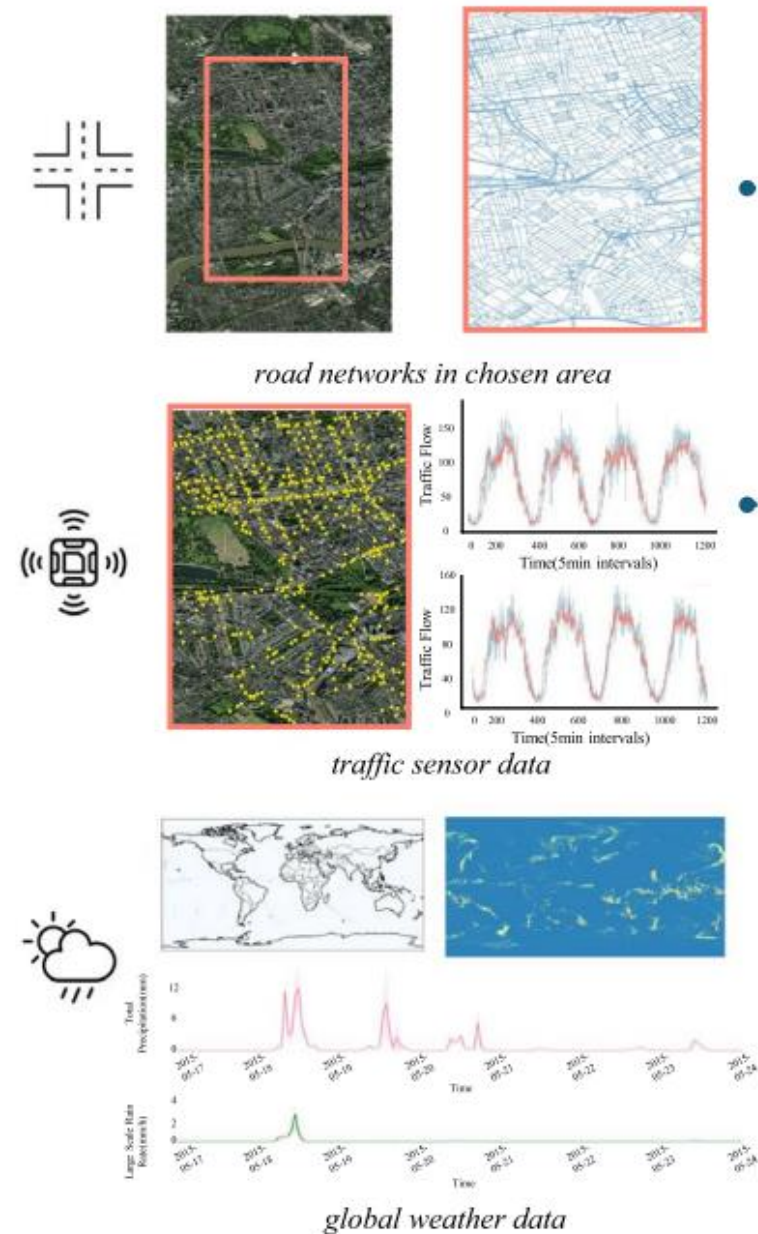
- **A Large-Scale, Multi-City Integrated Resource:** It provides an unprecedented, standardised collection of traffic, weather, and network data for 40 diverse global cities, specifically designed to enable robust comparative urban resilience studies.
- **A Novel and Replicable Harmonisation Framework:** We develop and implement a systematic methodology for the spatial and temporal harmonisation of point-based traffic sensor data, linear road networks, and gridded precipitation data, offering a transferable approach for future global urban data integration.
- **Ensuring Data Quality and Accessibility:** The dataset undergoes rigorous quality control procedures, including automated consistency checks and manual validation, to ensure reliability across all cities. The data is structured in accessible formats that support diverse analytical workflows while maintaining technical rigor, facilitating both academic research and practical applications in urban transportation management.
- **Providing Data Infrastructure for Advanced Impact and Resilience Analysis:** By providing deeply integrated and contextualised data, IUTF creates the essential empirical foundation needed to facilitate sophisticated investigations into how precipitation events affect urban mobility patterns and to support the future development and validation of quantitative resilience models.

Methods

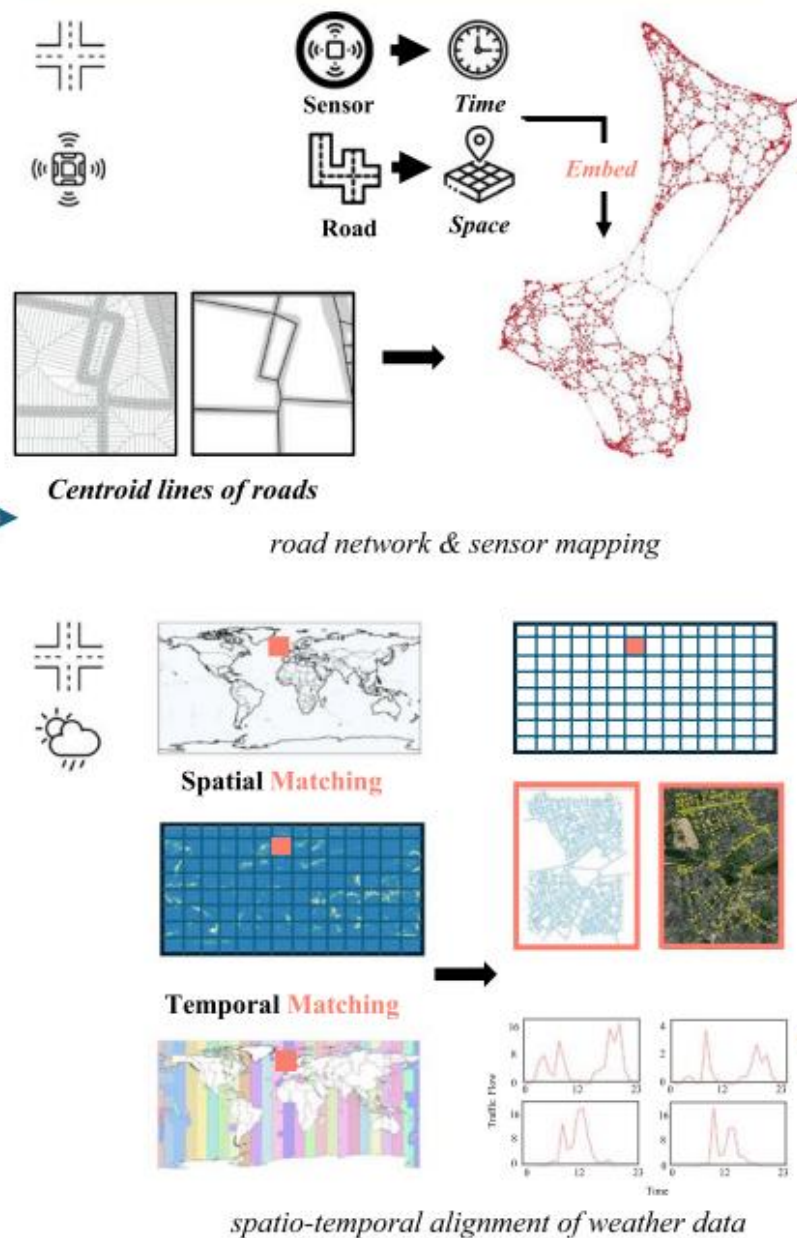
Three-Stage Pipeline

- Stage 1 (Data Source Acquisition and Preprocessing): Interfacing with **UTD19** (Traffic), **OSM** (Road Network), and **ERA5** (Meteorology).
- Stage 2 (Time and Space Alignment): Resolving cross-dimensional spatial alignment and temporal matching of 3D data from "points (sensors) - lines (road networks) - surfaces (weather grids)".
- Stage 3 (Data Compilation and Validation): Constructing a matrix and completing quality control and technical verification.

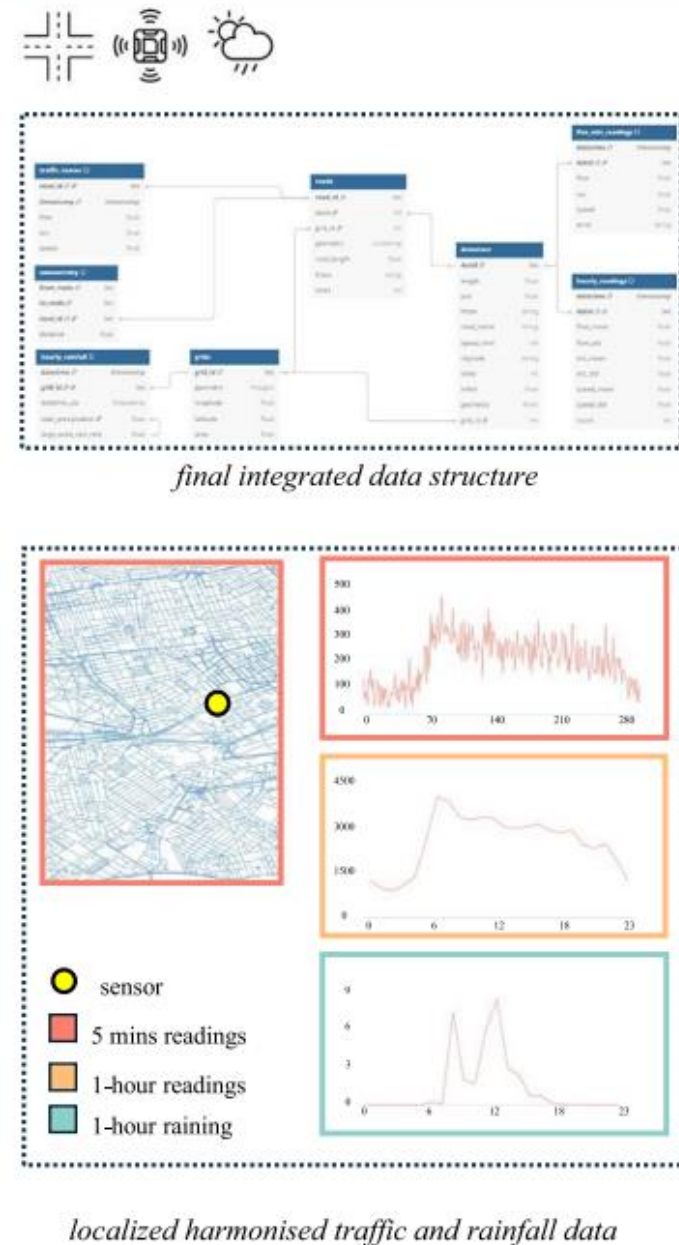
(a) Open Data



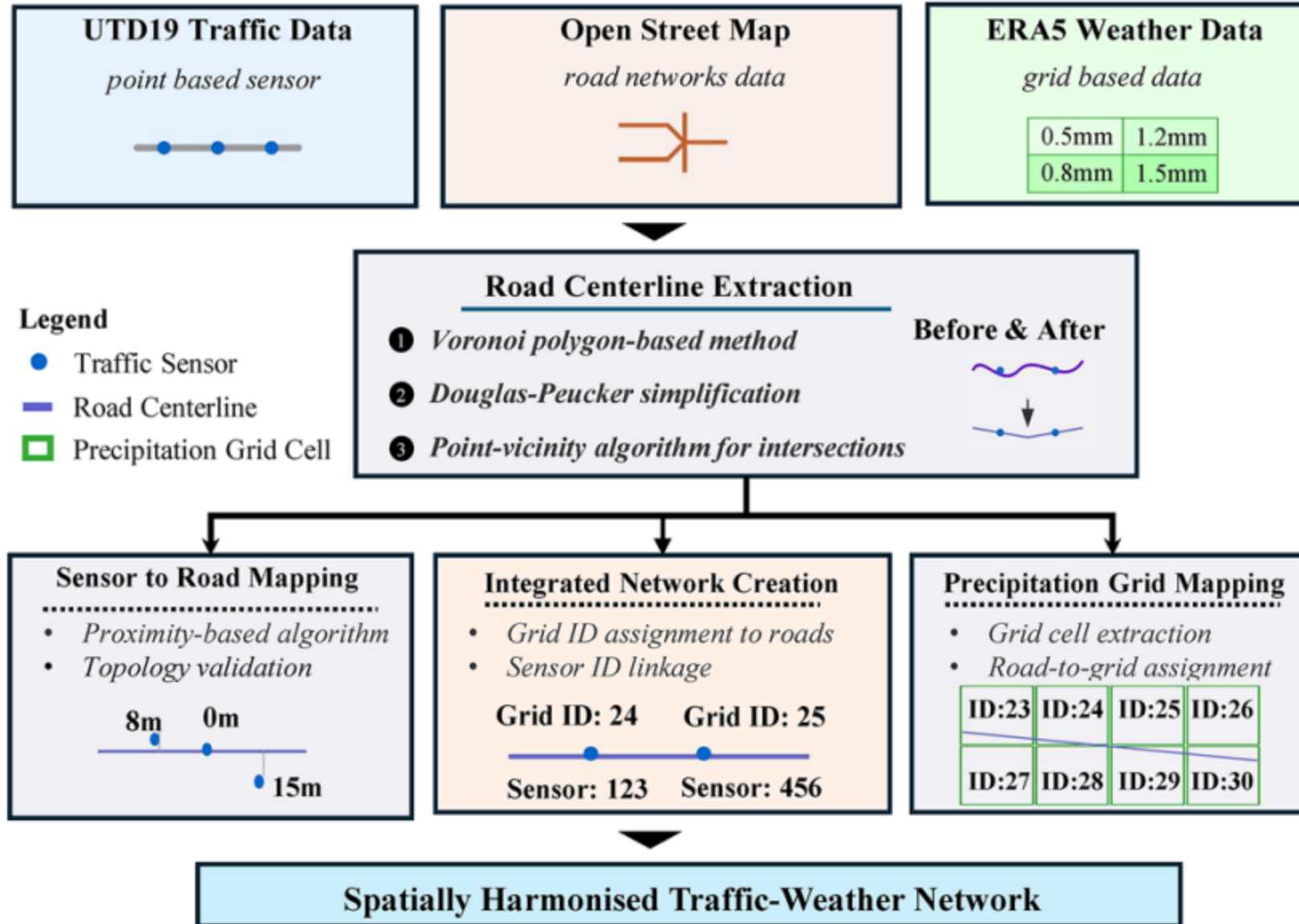
(b) Spatial & Temporal Harmonisation & Integration



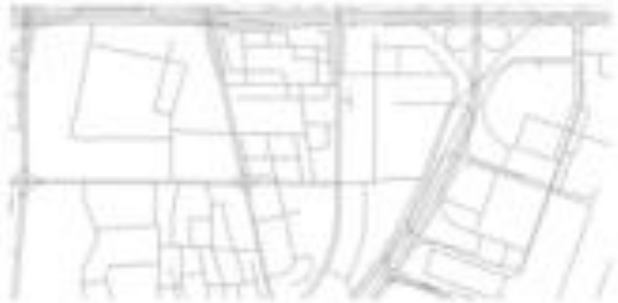
(c) Dataset



(a) Spatial Harmonisation Process



(b)



1. original road network



2. buffered road polygons



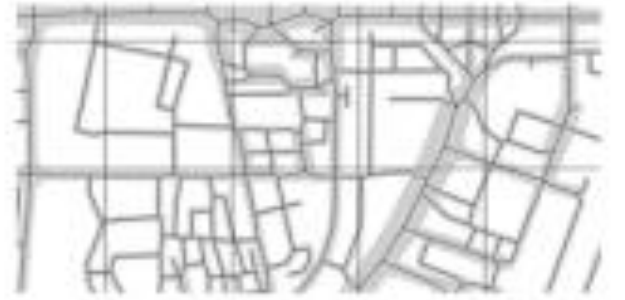
3. voronoi tessellation & grid



4. initial centreline extraction



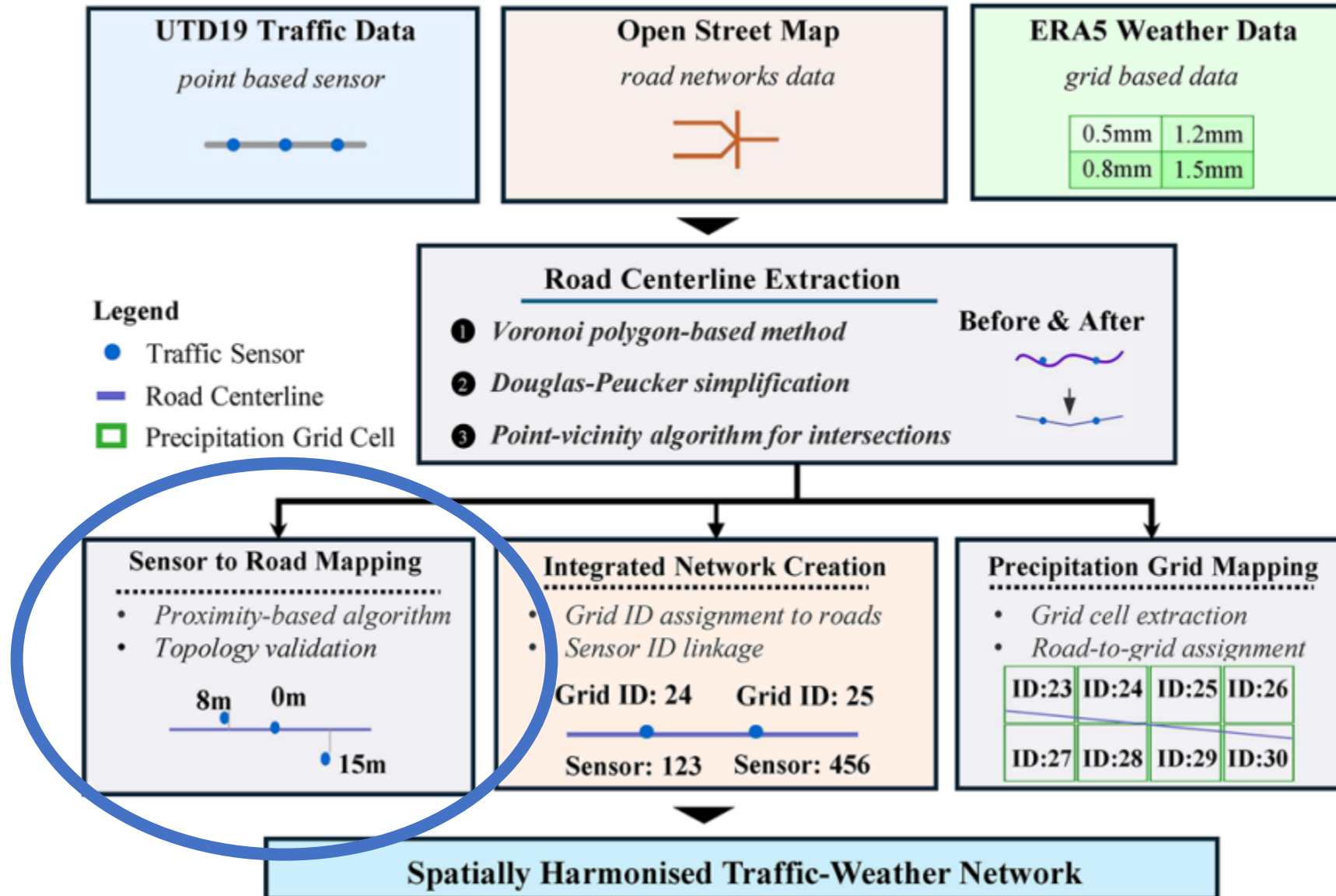
5. simplified & cleaned centrelines



6. final road centrelines

(a)

Spatial Harmonisation Process



Precipitation data integration. The integration of meteorological data with transportation networks presents a significant methodological challenge due to fundamental differences in spatial representation and resolution between these data sources. ERA5 reanalysis provides precipitation data at approximately 31 km spatial resolution in a regular grid format, while road networks consist of linear features with highly variable spatial density and geometric complexity. This spatial scale mismatch creates substantial difficulties for establishing meaningful relationships between weather conditions and traffic patterns, as a single ERA5 grid cell may encompass multiple distinct road segments with potentially different precipitation exposure characteristics. Thus, this stage addresses the fundamental challenge of accurately aligning gridded meteorological data with fine-scale transportation infrastructure.

Spatial overlay and grid-road intersection analysis. The first stage, Spatial Overlay and Grid-Road Intersection Analysis, begins by extracting only the relevant ERA5 grid cells that geometrically intersect with each city's road network boundary, creating a focused precipitation dataset for each urban area while significantly reducing computational demands. The process then employs computational geometry algorithms to determine precise intersection relationships between linear road segments and rectangular grid cells. This analysis requires coordinate system harmonisation, as road networks typically use local projected coordinate systems while ERA5 data employs WGS84 geographic coordinates, necessitating appropriate datum transformations to maintain spatial accuracy. For each road segment, the algorithm calculates the exact geometric relationship with intersecting grid cells, including the proportional length of road segment within each grid cell and the total intersection area, forming the foundation for subsequent precipitation attribution calculations.

Dual attribution methodology implementation. The second stage implements a dual attribution methodology specifically designed to support different analytical requirements through parallel processing pathways. The detailed attribution pathway preserves individual precipitation time series for each grid cell that intersects with a road segment, maintaining these as indexed data structures that retain the full spatial granularity of ERA5 data and enable analysis of precipitation variability along extended road segments. Simultaneously, the aggregated attribution pathway creates single precipitation values for each road segment through area-weighted averaging, calculated using the formula:

$$P_{segment} = \frac{\sum_{i=1}^n P_{grid_i} \times L_i}{\sum_{i=1}^n L_i} \quad (1)$$

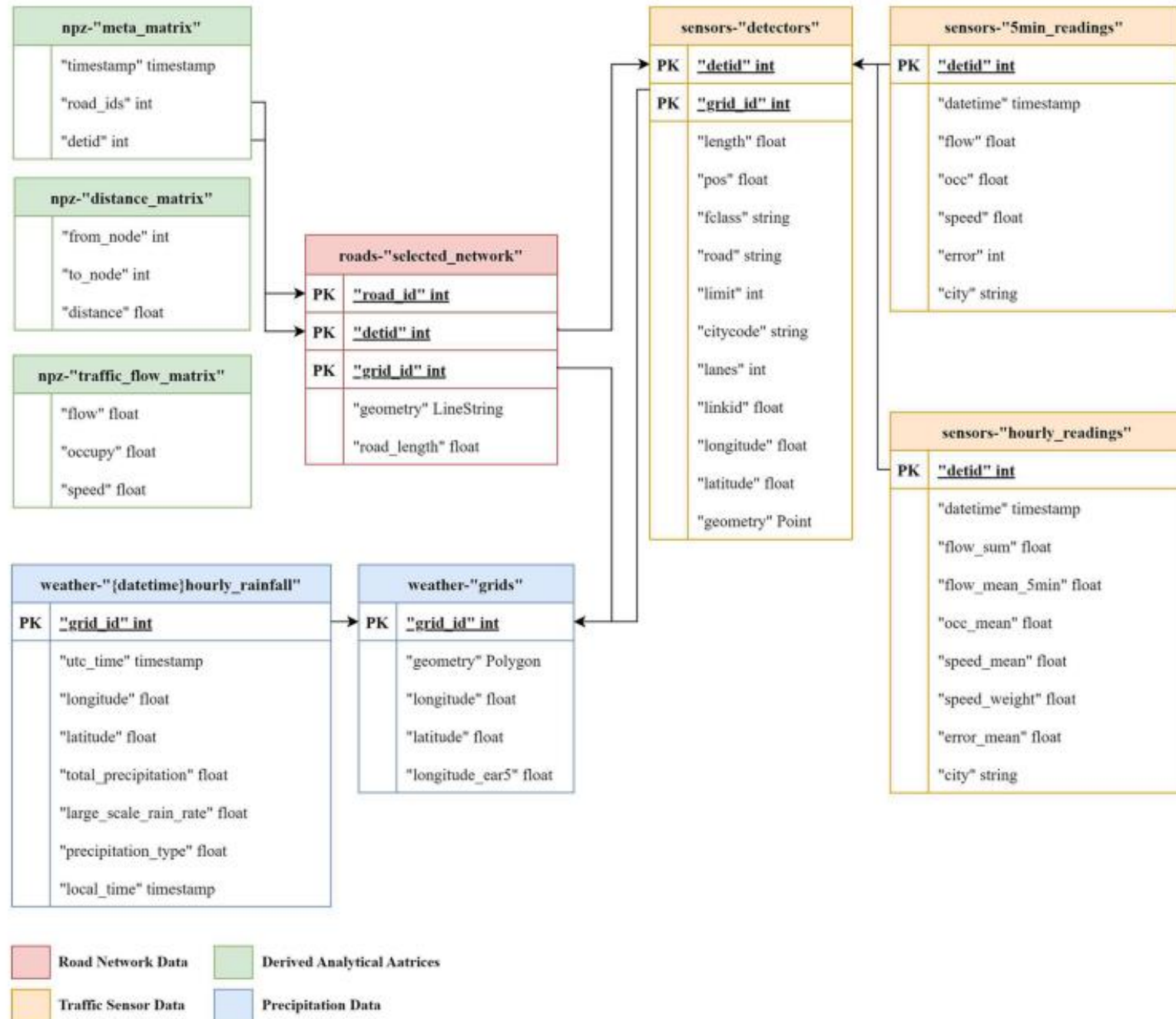
where $P_{segment}$ represents the final precipitation value for the road segment, P_{grid_i} is the precipitation value in grid cell i , L_i is the length of road segment within grid cell i , and n is the total number of grid cells that intersect with the road segment. This weighted averaging approach ensures that grid cells containing longer portions of a road segment contribute proportionally more to the final precipitation attribution, accounting for the varying spatial influence of different grid cells on individual road segments.

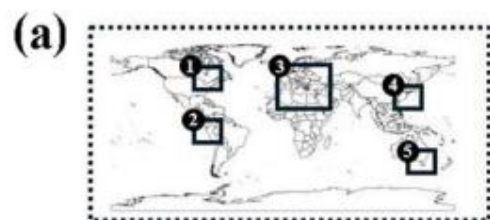
Temporal resolution aggregation.

$$Flow_{hourly} = \sum_{j=1}^{12} Flow_{5min_j}$$

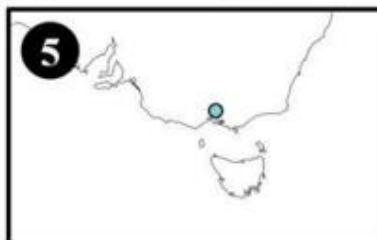
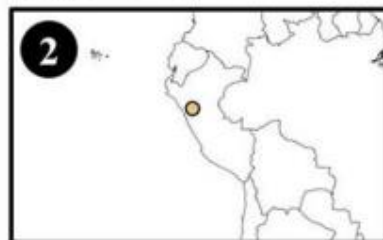
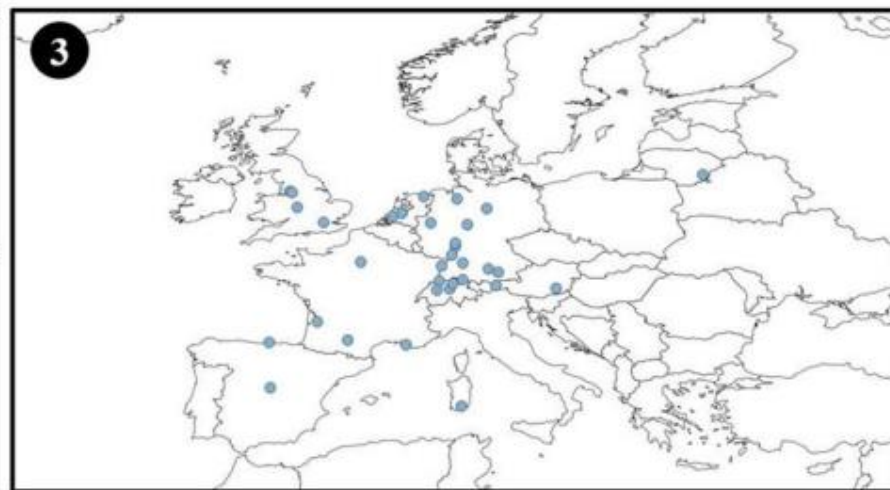
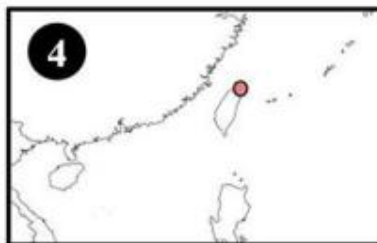
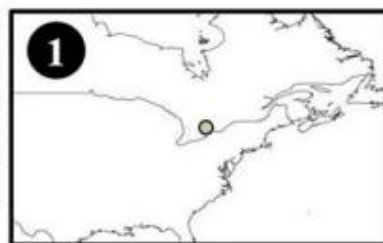
$$Speed_{hourly} = \frac{\sum_{j=1}^{12} (Speed_{5min_j} \times Flow_{5min_j})}{\sum_{j=1}^{12} Flow_{5min_j}}$$

$$Occupancy_{hourly} = \frac{1}{12} \sum_{j=1}^{12} Occupancy_{5min_j}$$



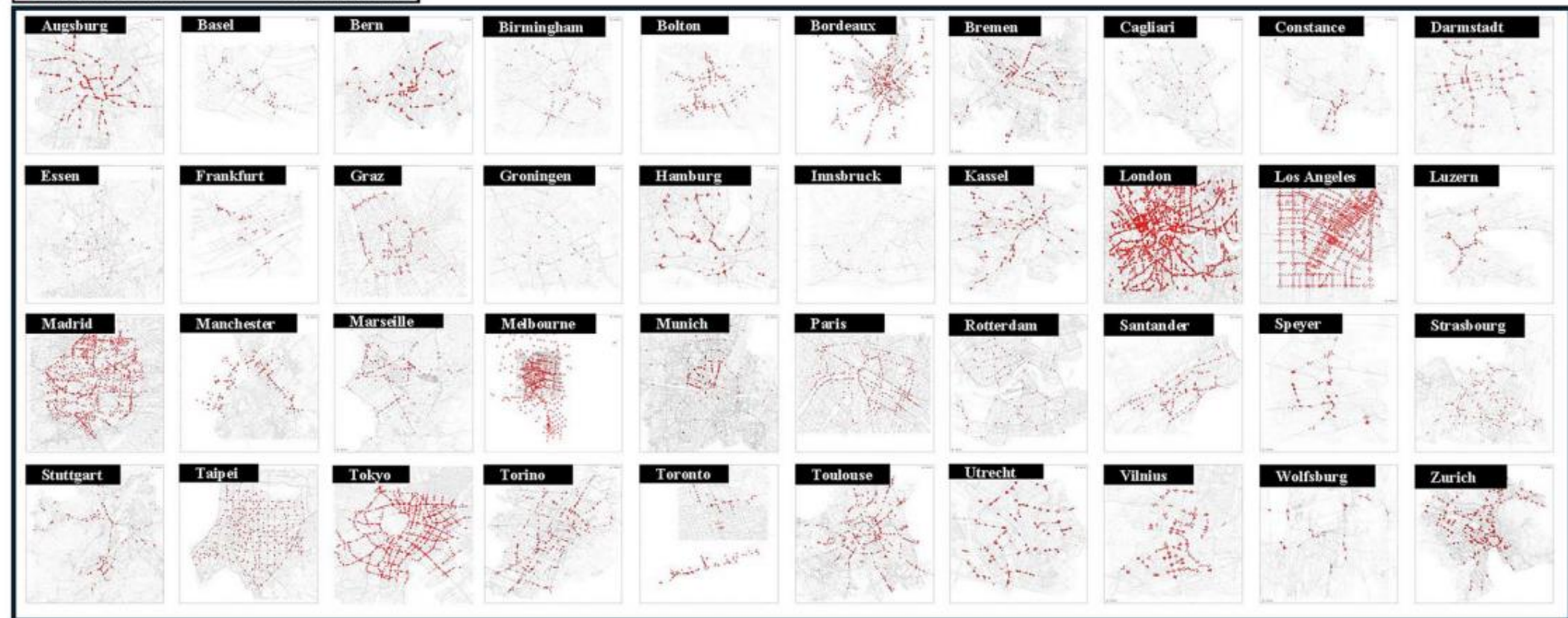


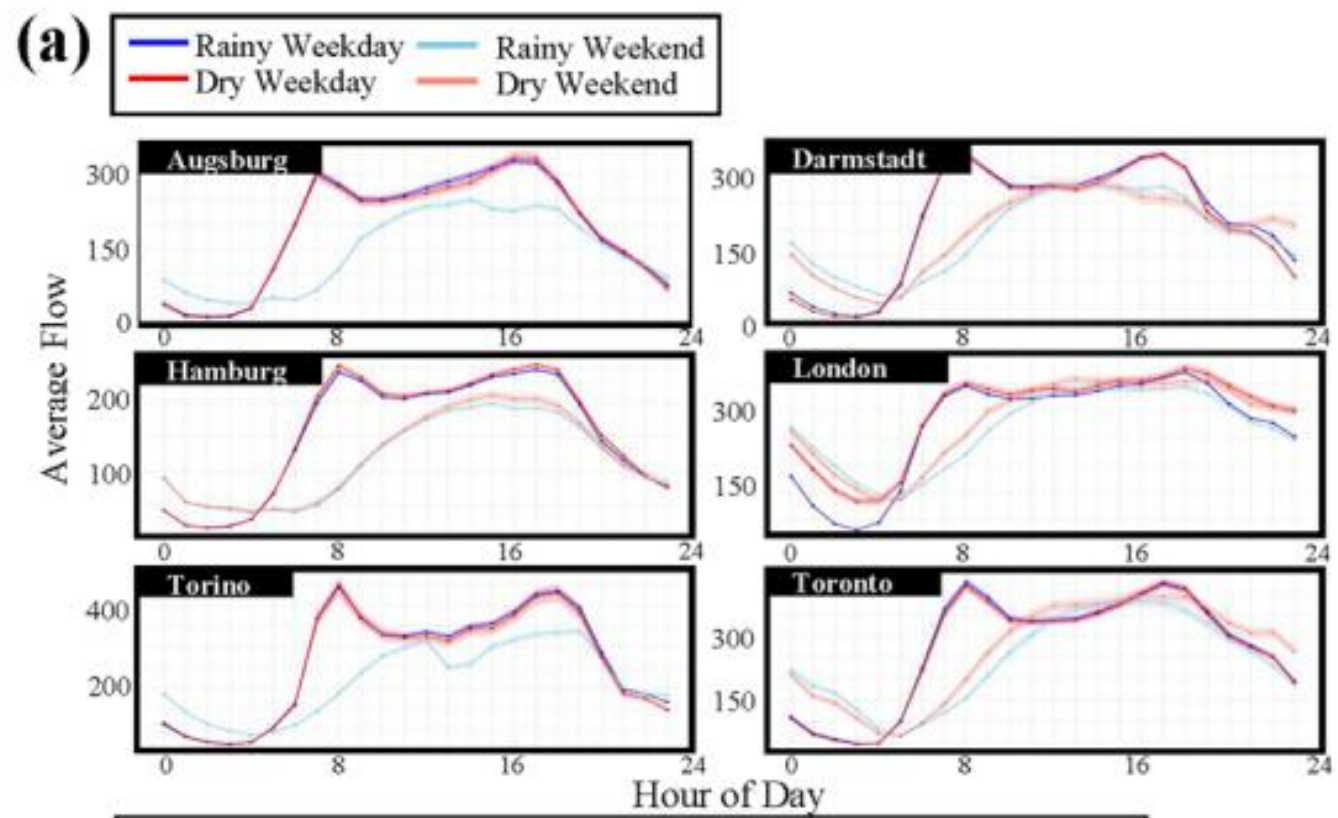
- Asia
- Australia
- Europe
- North America
- South America

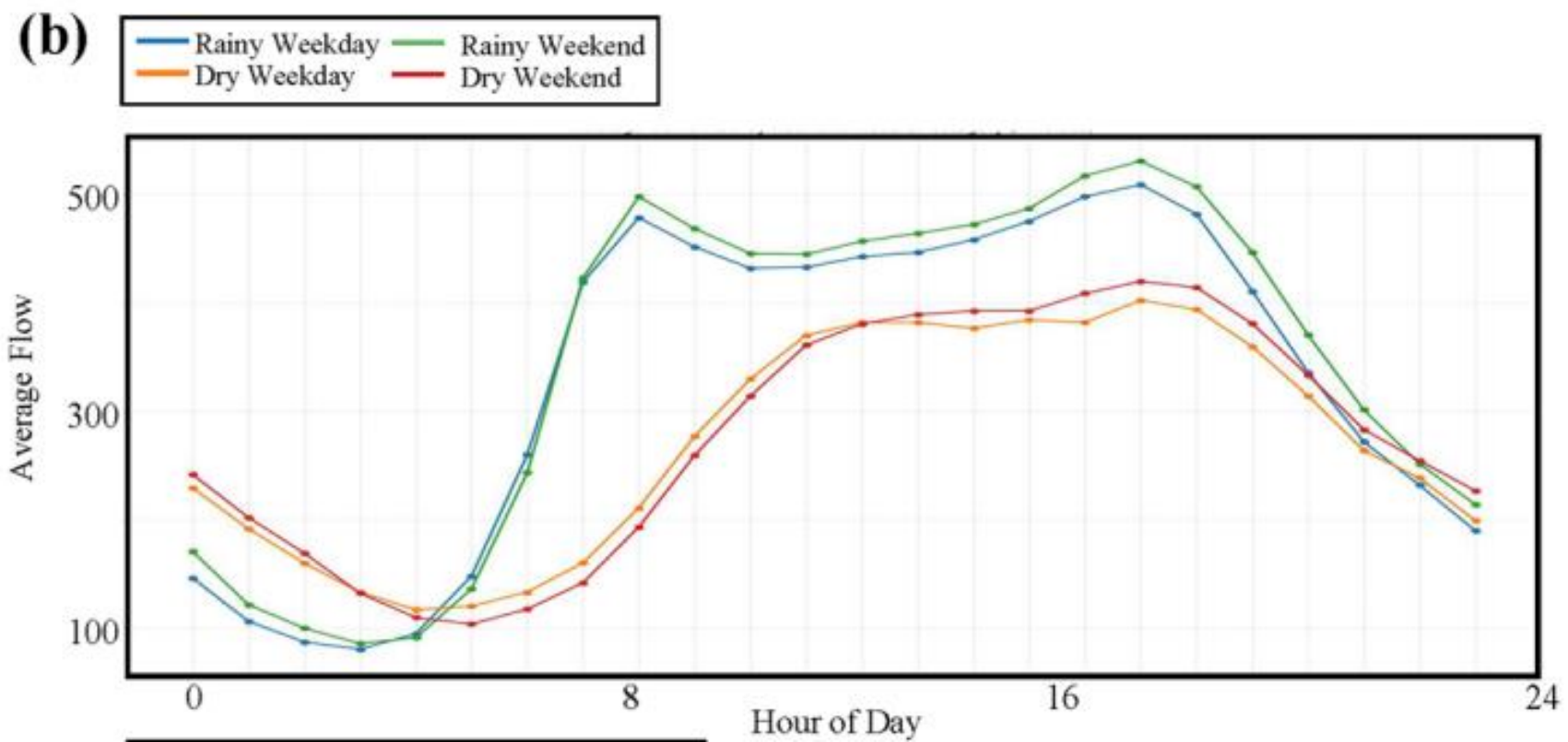


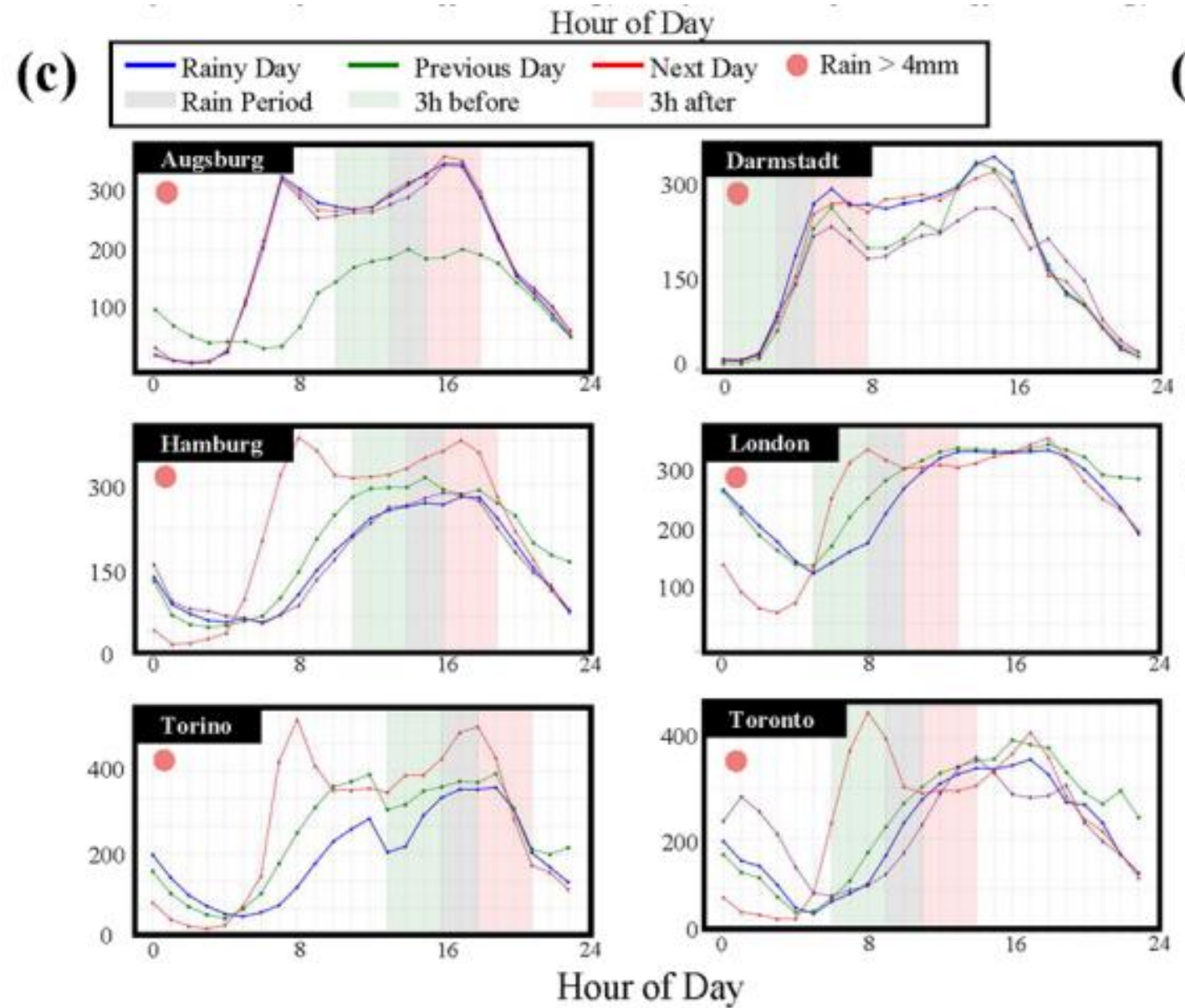
(b)

● Sensor location — Road networks

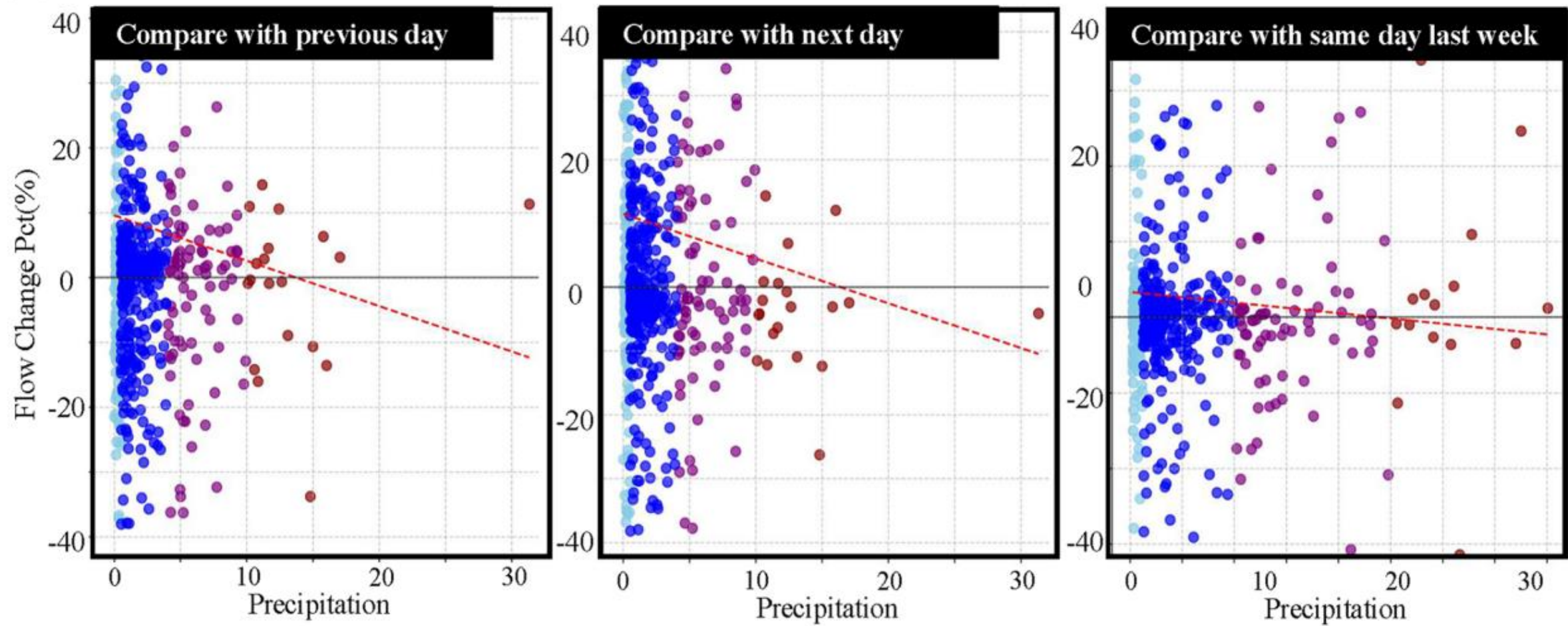




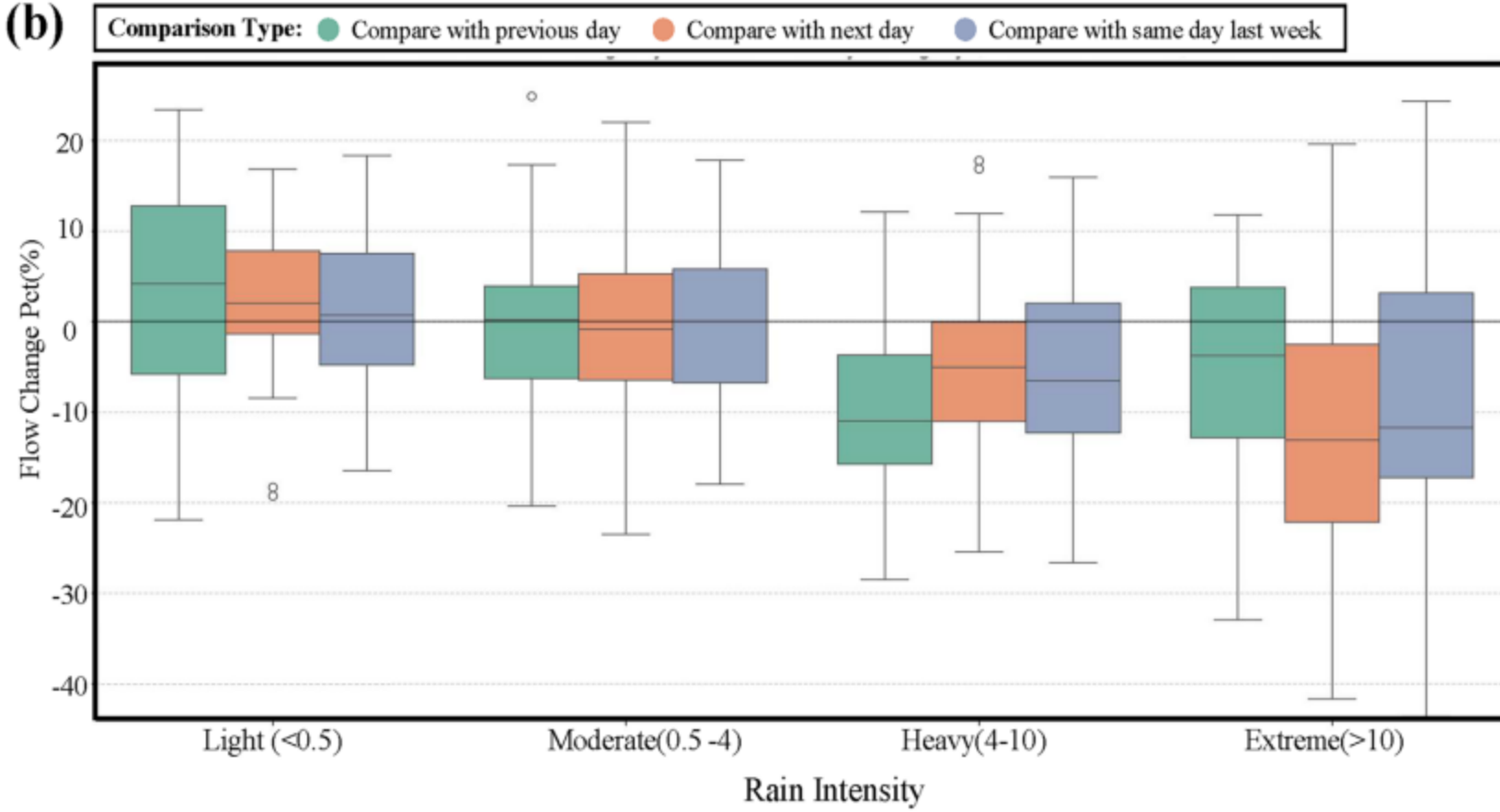




(a) Rainfall Intensity(mm/h): ● Light (< 0.5) ● Moderate (0.5 - 4) ● Heavy (4 - 10) ● Extreme (> 10) - - - Trend



(b)





Book Introduction: Traffic Flow Dynamics Data, Models and Simulation

Chapter.18 part1:Traffic Flow Breakdown and Traffic-State Recognition



Three Factors of Traffic Breakdown

1. High Traffic Flow: Road demand is close to or reaches road capacity, and is in a critical state.
2. Bottlenecks: Physical bottlenecks (such as lane reduction, ramp merging, curves and ramps) or behavioral bottlenecks (such as accidents, lane change interference) that lead to a decrease in local traffic capacity.
3. Disturbances : Sudden braking of individual vehicles, unannounced lane changes, or driver reaction delays—these micro-disturbances can be amplified under critically high flow conditions, triggering "phantom traffic jams."

In the next two sections, we summarize the stylized facts of the spatiotemporal evolution of congested traffic patterns

Traffic Flow Breakdown: Three Ingredients to Make a Traffic Jam

- Using specialized methods such as the **Adaptive Smoothing Method**, the spatiotemporal dynamic evolution of congested areas was reconstructed.
- Analyzing these data, we observe that nearly all real-world traffic breakdowns are caused by the simultaneous action of three factors: **High traffic load**, a **bottleneck**, and **disturbances of traffic flow** caused by individual drivers. In the following, we describe each of these three “ingredients to make a traffic jam”

Traffic Flow Breakdown: Three Ingredients to Make a Traffic Jam

High traffic load.

This factor is the most obvious one: If the traffic load on the network is low enough then disturbances caused by bottlenecks or abrupt driving maneuvers cannot grow and propagate since traffic is unconditionally stable for sufficiently low densities

Bottlenecks.

Phantom traffic jams do not occur on real road networks which are necessarily inhomogeneous. We define a bottleneck as a local reduction of the road capacity . Bottlenecks can be permanent attributes of the infrastructure (e.g., on-ramps, off-ramps, roadworks, etc.) or temporary, e.g., when caused by accidents.

Traffic Flow Breakdown: Three Ingredients to Make a Traffic Jam

Bottlenecks come in many forms. The most common instances on highways and major roadways are the following¹:

- On-ramps and off-ramps.
- Lane closures, road narrowings and curves (particularly at roadworks and road construction sites).
- Uphill and downhill gradients.
- Temporary obstructions caused by accidents (including complete road closures).
- spectacular accidents or jammed traffic on the opposite side of the road.

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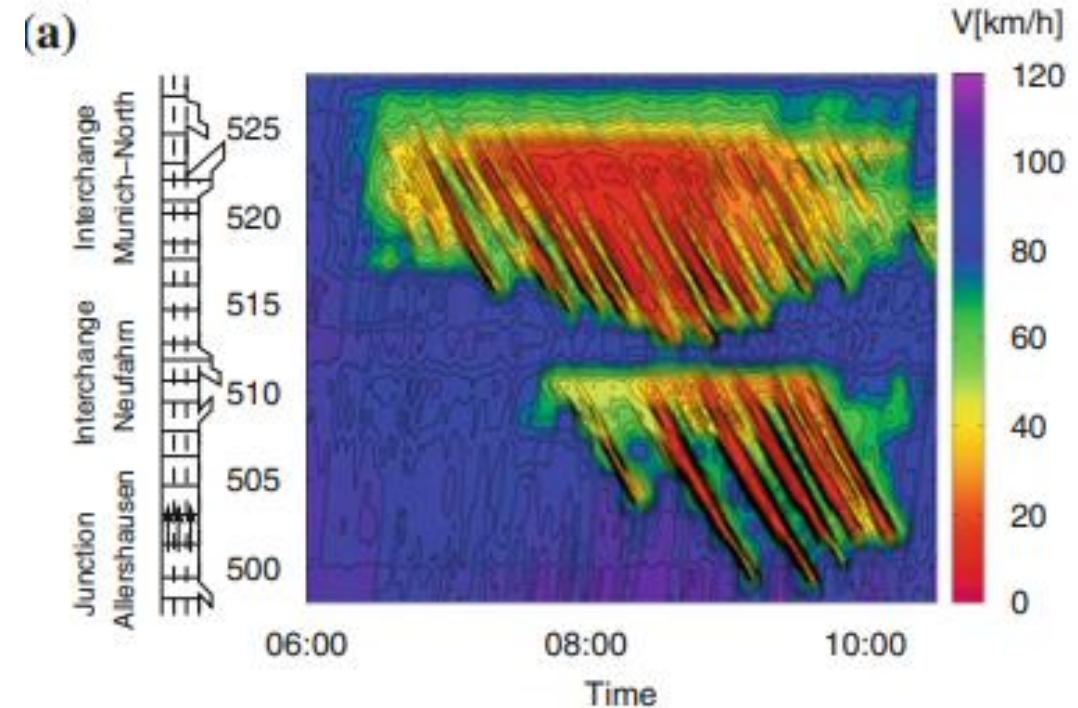
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Traffic Flow Breakdown: Three Ingredients to Make a Traffic Jam

When traffic breaks down at the bottleneck and congested traffic has formed upstream of it, one says that the bottleneck is activated. Typically, bottleneck activation is accompanied by a drop in the traffic flow through the bottleneck, i.e., the bottleneck capacity, of the order of 10–20%. This capacity drop from the static to the dynamic bottleneck capacity is also observed in simulations of microscopic and second-order macroscopic models

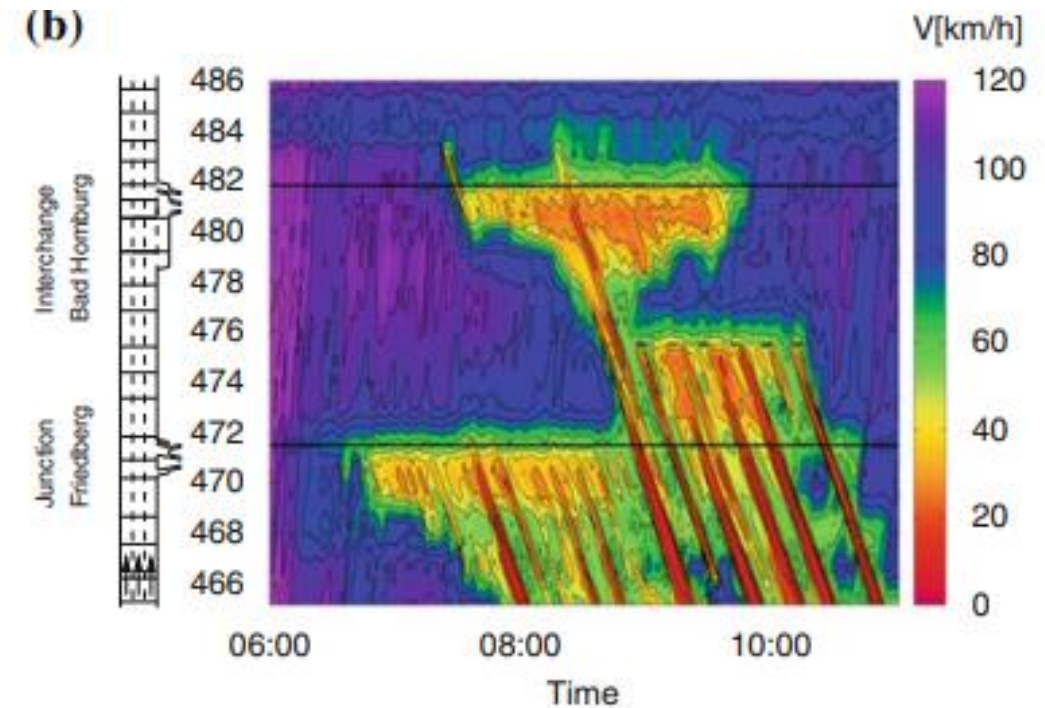
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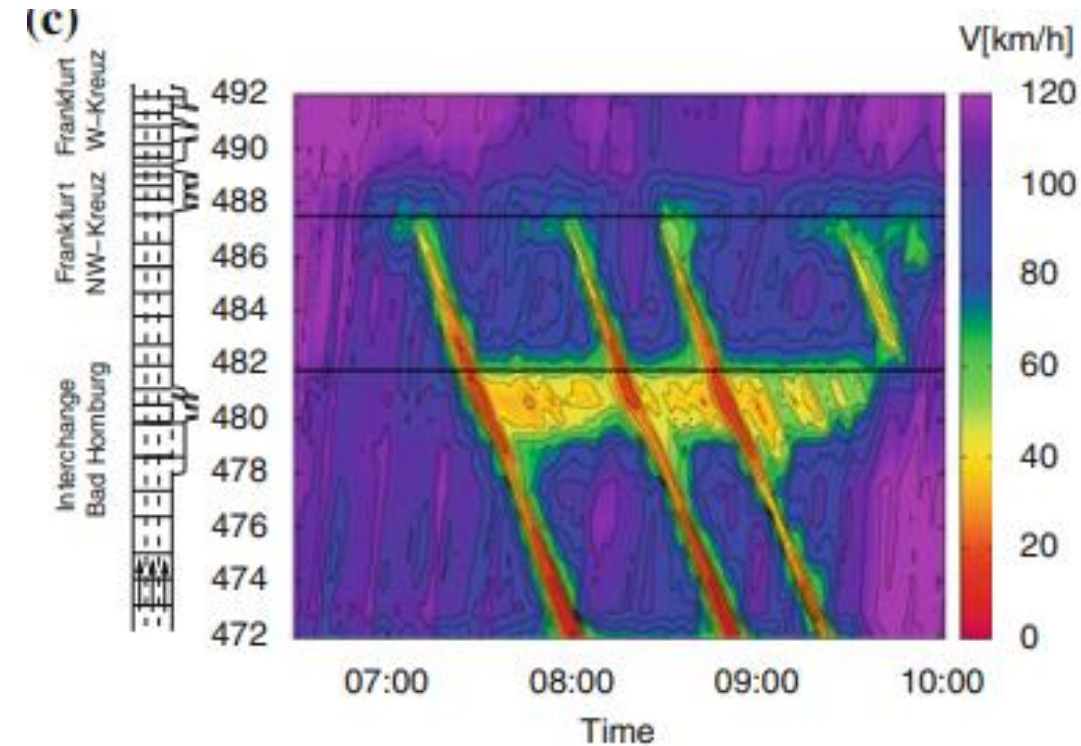
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Traffic Flow Breakdown: Three Ingredients to Make a Traffic Jam

Disturbances caused by individual drivers

The third factor that is necessary for traffic breakdowns consists of perturbations in the traffic flow itself. While, theoretically, an infinitesimal perturbation suffices if traffic flow is linearly unstable, this does not apply to real traffic flow where non-congested traffic is, at most, metastable. Disturbances in the traffic flow are realized, e.g., by inattentive drivers braking abruptly, by speeding cars, lane changes, or by **elephant races**, i.e., trucks overtaking each other at speed differences, particularly on roads with only two lanes per direction